



Application of Taguchi Method of Experimental Design for Chemical Process Optimisation: A Case Study

Purpose: The purpose of this practical assignment was to reproduce the Taguchi experimental design and analysis presented by Patyal et al. (2013) using the Isalos Analytics Platform. The study investigates four control factors: Temperature, Borex, pH, and Time. Two responses were evaluated: Loss on Ignition (LoI), for which lower values are preferred, and Borate (B₂O₃), for which higher values are preferred.

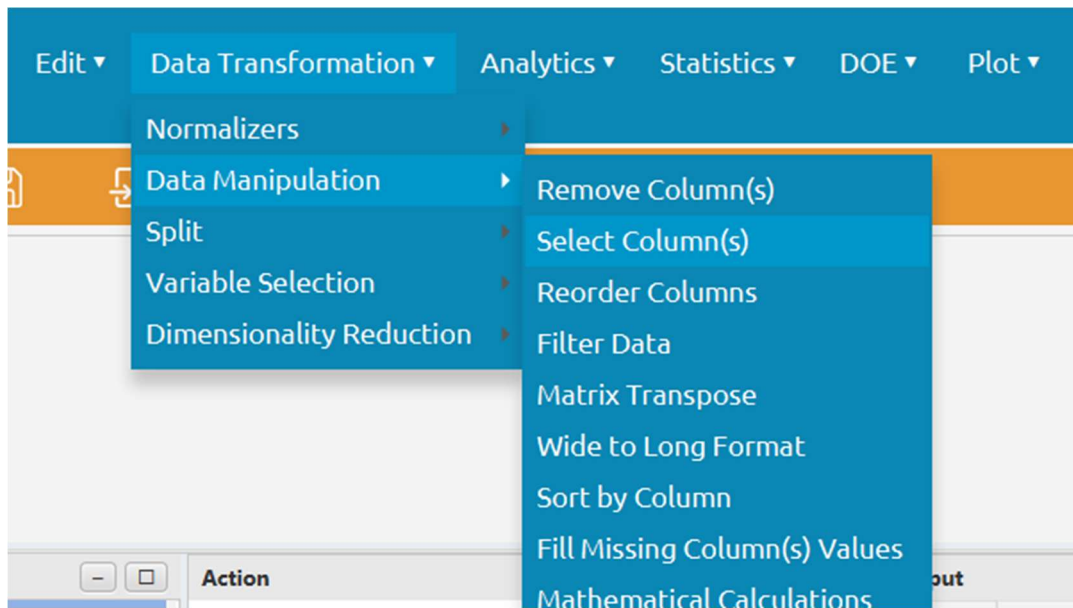
Isalos version used: 2.2.0

Scientific article: Patyal et al. (2013) – Application of Taguchi Method of Experimental Design for Chemical Process Optimisation: A Case Study

Step 1: Taguchi Design

Rename the first tab from “Action” to “Factors”. Enter the nine experimental runs reported in Table 3 of the paper using the four control factors: Temperature, Borex, pH, and Time. Add a dummy Noise Factor column, as a noise factor is required by the Isalos Robust Parameter functionality.

Col1	Col2 (I)	Col3 (I)	Col4 (D)	Col5 (I)	Col6 (I)
User Row ID	Temp	Borex	pH	Time	Noise_Factor
	60	1	1.5	6	1
	60	3	2.5	10	2
	60	6	3	15	1
	80	1	2.5	15	2
	80	3	3	6	1
	80	6	1.5	10	2
	100	1	3	10	1
	100	3	1.5	15	2
	100	6	2.5	6	1



Results (right spreadsheet) :

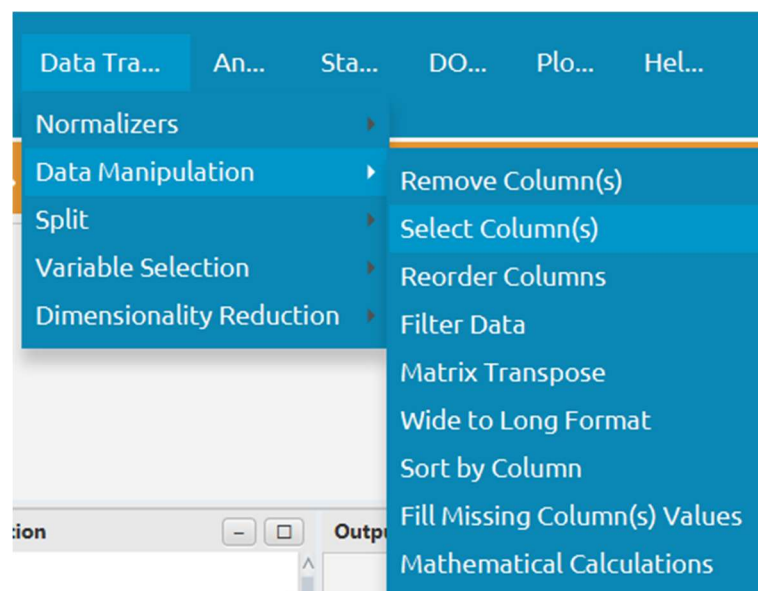
Output							
	Col1	Col2 (I)	Col3 (I)	Col4 (D)	Col5 (I)	Col6 (I)	
User Header	User Row ID	Temp	Borex	pH	Time	Noise_Factor	
1		60	1	1.5	6	1	
2		60	3	2.5	10	2	
3		60	6	3	15	1	
4		80	1	2.5	15	2	
5		80	3	3	6	1	
6		80	6	1.5	10	2	
7		100	1	3	10	1	
8		100	3	1.5	15	2	
9		100	6	2.5	6	1	
10							

Step 2: Definition of Response Variables

Create a new spreadsheet and rename it “Responses”. Enter the experimental response values reported in Table 4 of the paper for the nine experimental runs. The two responses are Loss on Ignition (LoI) and Borate (B2O3), with three replicate measurements for each response: LoI_R1, LoI_R2, LoI_R3, B2O3_R1, B2O3_R2, and B2O3_R3.

After entering the data, go to Data Transformation → Data Manipulation → Select Column(s) and select all six response columns. Click Execute to complete the response dataset.

Col1	Col2 (D)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	Col7 (D)
User Row ID	Lol_R1	Lol_R2	Lol_R3	B2O3_R1	B2O3_R2	B2O3_R3
	25.55	25.66	25.77	42.29	42.19	42.09
	30.38	30.48	30.28	30.8	34.8	32.8
	25.76	25.36	25.56	41.25	41.15	41.35
	19.9	19.95	20	38.72	39.32	39.02
	20.58	20.08	20.33	39.82	39.72	39.62
	19.22	19.02	19.42	30.8	38.8	34.8
	15.34	15.24	15.14	42.3	42.15	42
	14.03	14.18	14.33	45.35	45.3	45.4
	12.9	12.6	12.3	46.63	46.43	46.23



Select Column(s)

☒ Select Columns Manually
☐ Select Matching Columns from Node
☐ Select Columns Manually and from Node

Select Node

Excluded Columns

Included Columns

Col2 -- Lol_R1
 Col3 -- Lol_R2
 Col4 -- Lol_R3
 Col5 -- B2O3_R1
 Col6 -- B2O3_R2
 Col7 -- B2O3_R3

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Execute

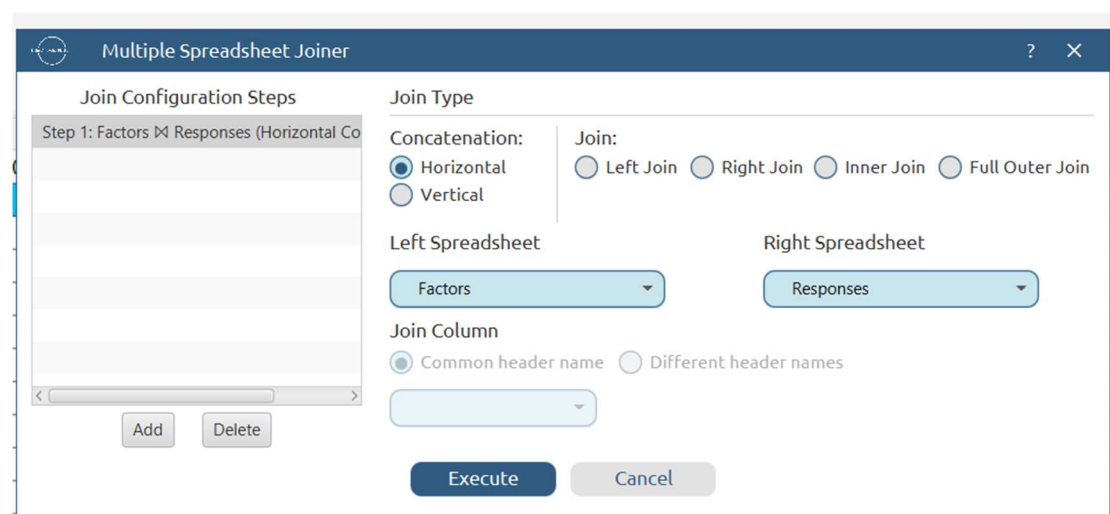
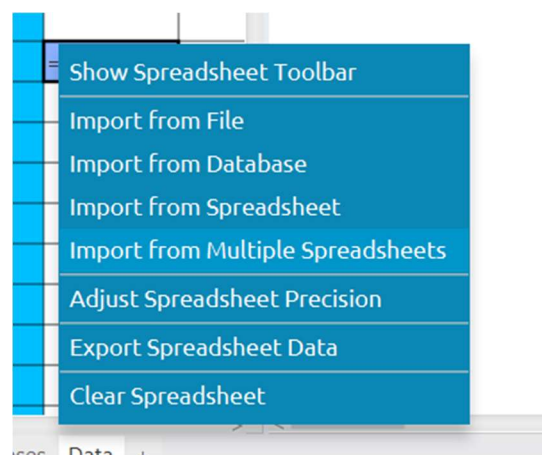
Cancel

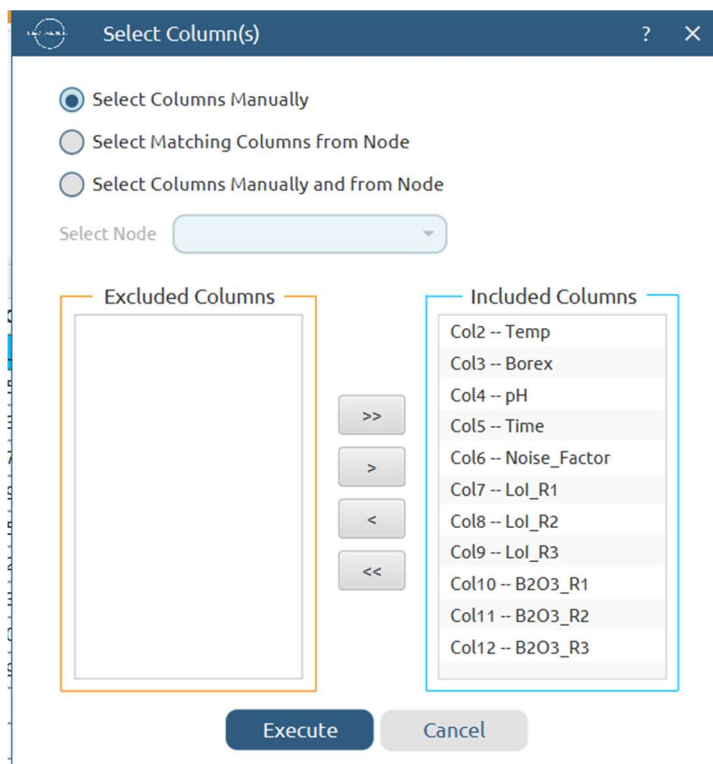
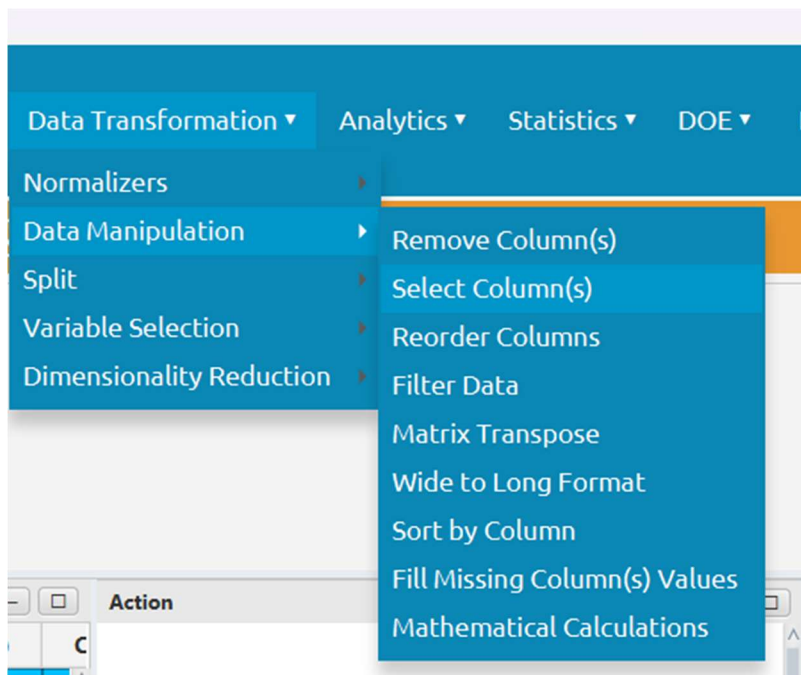
Output								
	Col1	Col2 (D)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	Col7 (D)	Col8
User Header	User Row ID	Lol_R1	Lol_R2	Lol_R3	B2O3_R1	B2O3_R2	B2O3_R3	
1		25.55	25.66	25.77	42.29	42.19	42.09	
2		30.38	30.48	30.28	30.8	34.8	32.8	
3		25.76	25.36	25.56	41.25	41.15	41.35	
4		19.9	19.95	20	38.72	39.32	39.02	
5		20.58	20.08	20.33	39.82	39.72	39.62	
6		19.22	19.02	19.42	30.8	38.8	34.8	
7		15.34	15.24	15.14	42.3	42.15	42	
8		14.03	14.18	14.33	45.35	45.3	45.4	
9		12.9	12.6	12.3	46.63	46.43	46.23	
10								
11								

Step 3: Data Preparation

Create a new spreadsheet and rename it “Data”. To combine the Taguchi design with the Response data, right-click on the spreadsheet and select Import from Multiple Spreadsheets. In the Multiple Spreadsheet Joiner, select Horizontal concatenation, with Taguchi Design as the Left Spreadsheet and Responses as the Right Spreadsheet. Click Execute.

Then, go to Data Transformation → Data Manipulation → Select Column(s). Select the four control factors Temp, Borex, pH, and Time, the dummy Noise_Factor, and the six response columns Lol_R1, Lol_R2, Lol_R3, B2O3_R1, B2O3_R2, and B2O3_R3. Click Execute to complete the final dataset.





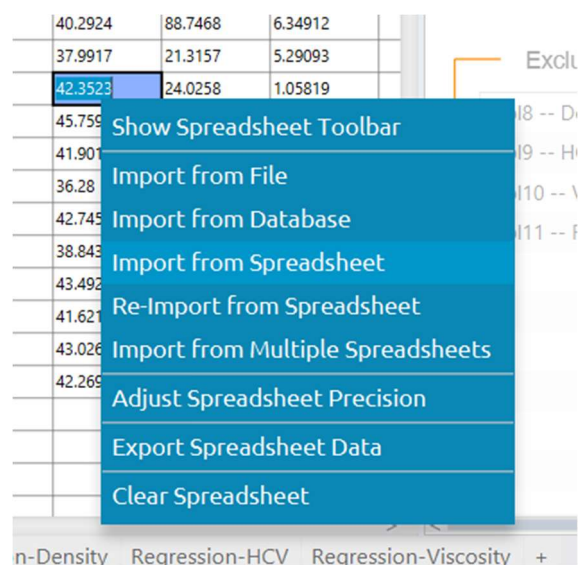
Results :

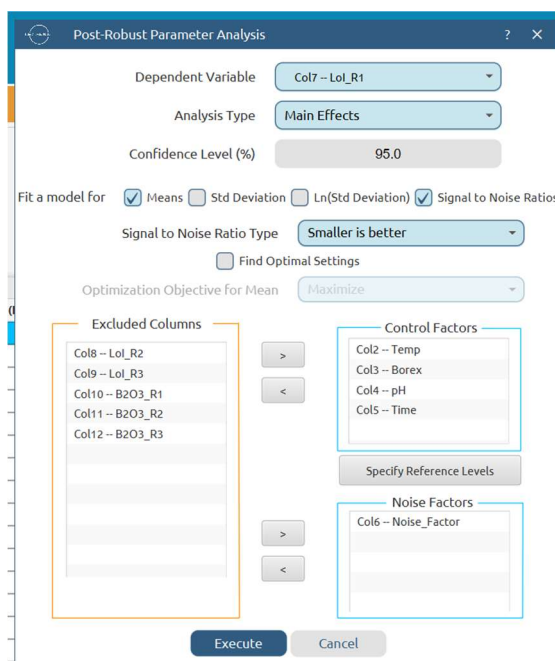
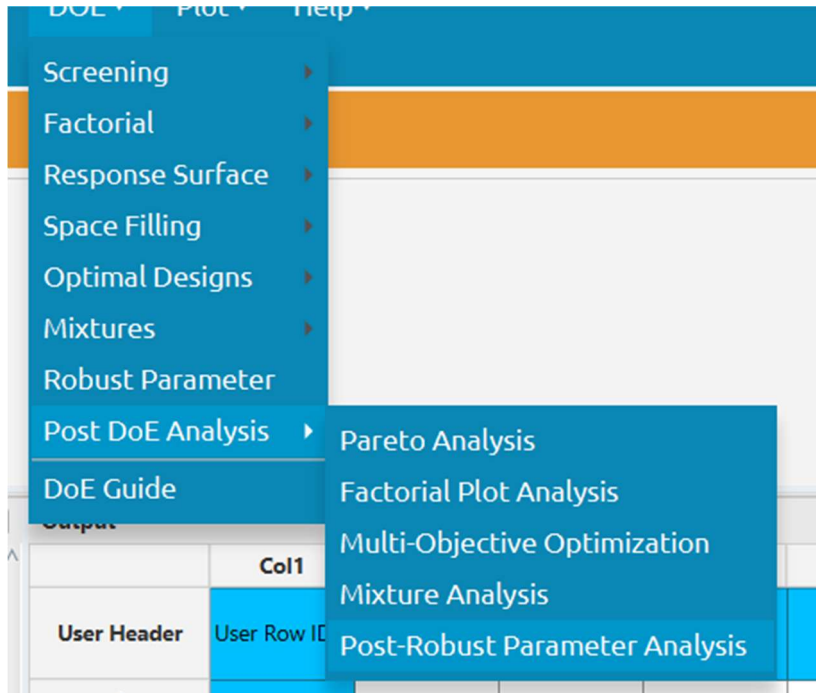
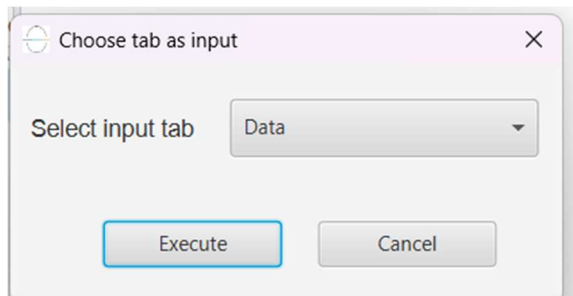
Output												
	Col1	Col2 (I)	Col3 (I)	Col4 (D)	Col5 (I)	Col6 (I)	Col7 (D)	Col8 (D)	Col9 (D)	Col10 (D)	Col11 (D)	Col12 (D)
User Header	User Row ID	Temp	Borex	pH	Time	Noise_Factor	LoI_R1	LoI_R2	LoI_R3	B2O3_R1	B2O3_R2	B2O3_R3
1		60	1	1.5	6	1	25.55	25.66	25.77	42.29	42.19	42.09
2		60	3	2.5	10	2	30.38	30.48	30.28	30.8	34.8	32.8
3		60	6	3	15	1	25.76	25.36	25.56	41.25	41.15	41.35
4		80	1	2.5	15	2	19.9	19.95	20	38.72	39.32	39.02
5		80	3	3	6	1	20.58	20.08	20.33	39.82	39.72	39.62
6		80	6	1.5	10	2	19.22	19.02	19.42	30.8	38.8	34.8
7		100	1	3	10	1	15.34	15.24	15.14	42.3	42.15	42
8		100	3	1.5	15	2	14.03	14.18	14.33	45.35	45.3	45.4
9		100	6	2.5	6	1	12.9	12.6	12.3	46.63	46.43	46.23
10												

Step 4: LoI_1 Post-Robust Parameter Analysis

Create a new spreadsheet and rename it “Taguchi LoI_R1”. Right-click on the spreadsheet, select Import from Spreadsheet, choose the Data tab as input, and click Execute.

Then, go to DOE → Post DOE Analysis → Post-Robust Parameter Analysis. Select LoI_R1 as the dependent variable and Main Effects as the analysis type. Select Means and Signal to Noise Ratios, and choose Smaller is better as the Signal-to-Noise Ratio type. Select Temp, Borex, pH, and Time as the Control Factors and Noise_Factor as the Noise Factor. Click Execute to obtain the main-effects and Signal-to-Noise Ratio results. Then, click Show Chart to visualize the results, including the Main Effects and Signal-to-Noise Ratio plots.





Results :

Output																		
	Col1	Col2 (S)	Col3 (S)	Col4 (S)	Col5 (S)	Col6 (S)	Col7 (S)	Col8 (S)	Col9 (S)	Col10 (S)	Col11	Col12 (S)	Col13 (S)	Col14 (S)	Col15 (S)	Col16 (S)	Col17 (S)	Col18 (S)
User Header	User Row ID	Fitted Model for the Means of Lol_R1										Fitted Model for the Signal To Noise Ratio of Lol_R1						
1																		
2																		
3		Mean Model Response Table										Signal To Noise Ratio Model Response Table						
4		Level	Temp	Borex	pH	Time						Level	Temp	Borex	pH	Time		
5		1	27.23	20.2633333	19.6000000	19.6766667						1	-28.6728302	-25.9471289	-25.5880131	-25.5428399		
6		2	19.9	21.6633333	21.06	21.6466667						2	-25.9736789	-26.2872721	-25.9468704	-26.3477767		
7		3	14.0900000	19.2933333	20.5600000	19.8966667						3	-22.9564849	-25.3685930	-26.0681106	-25.7123774		
8		Delta	13.1400000	2.3700000	1.4600000	1.9700000						Delta	5.7163453	0.9186791	0.4800975	0.8049368		
9		Rank	1.0	2.0	4.0	3.0						Rank	1.0	2.0	4.0	3.0		
10																		
11		Mean Model ANOVA Table										Signal To Noise Ratio Model ANOVA Table						
12		Source	DF	Adj SS	Adj MS	F-Value	P-Value	Significant				Source	DF	Adj SS	Adj MS	F-Value	P-Value	Significant
13		Temp	2	260.1446000	130.0723000							Temp	2	49.0654806	24.5327403			
14		Borex	2	8.5178000	4.2589000							Borex	2	1.2943724	0.6471862			
15		pH	2	3.3032000	1.6516000							pH	2	0.3739714	0.1869857			
16		Time	2	6.9918000	3.4959000							Time	2	1.0803987	0.5401993			
17		Error	0									Error	0					
18		Total	8	278.9574000								Total	8	51.8142230				
19																		

Output																				
	Col1	Col2 (S)	Col3 (S)	Col4 (S)	Col5 (S)	Col6 (S)	Col7 (S)	Col8 (S)	Col9 (S)	Col10 (S)	Col11	Col12 (S)	Col13 (S)	Col14 (S)	Col15 (S)	Col16 (S)	Col17 (S)	Col18 (S)	Col19 (S)	Col20 (S)
21																				
22		Mean Model Coefficients Table										Signal To Noise Ratio Model Coefficients Table								
23																				
24		Variable	Coefficient	Std. Error	Lower CI	Upper CI	Test Statistic	df	P-Value	Significant		Variable	Coefficient	Std. Error	Lower CI	Upper CI	Test Statistic	df	P-Value	Significant
25		intercept	20.4066667	0.3333333	19.7533453	21.0599880		1.0				intercept	-25.8676647	0.3333333	-26.5209860	-25.2143433		1.0		
26		Temp_60.0	6.8233333	0.4714045	5.8993975	7.7472692		1.0				Temp_60.0	-2.8051655	0.4714045	-3.7291014	-1.8812297		1.0		
27		Temp_80.0	-0.5066667	0.4714045	-1.4306025	0.4172692		1.0				Temp_80.0	-0.1060142	0.4714045	-1.0299501	0.8179217		1.0		
28		Borex_1.0	-0.1433333	0.4714045	-1.0672692	0.7806025		1.0				Borex_1.0	-0.0794643	0.4714045	-1.0034001	0.8444716		1.0		
29		Borex_3.0	1.2566667	0.4714045	0.3327308	2.1806025		1.0				Borex_3.0	-0.4196074	0.4714045	-1.3435433	0.5043285		1.0		
30		pH_1.5	-0.8066667	0.4714045	-1.7306025	0.1172692		1.0				pH_1.5	0.2796516	0.4714045	-0.6442843	1.2035875		1.0		
31		pH_2.5	0.6533333	0.4714045	-0.2706025	1.5772692		1.0				pH_2.5	-0.0792057	0.4714045	-1.0031416	0.8447302		1.0		
32		Time_6.0	-0.7300000	0.4714045	-1.6539359	0.1939359		1.0				Time_6.0	0.3248248	0.4714045	-0.5991111	1.2487607		1.0		
33		Time_10.0	1.2400000	0.4714045	0.3160641	2.1639359		1.0				Time_10.0	-0.4801121	0.4714045	-1.4040480	0.4438238		1.0		

Step 5: Lol_2 and Lol_3 Post-Robust Parameter Analysis

Repeat the Post-Robust Parameter Analysis for the remaining Lol replicates. Create separate spreadsheets for Lol_R2 and Lol_R3, import the Data spreadsheet, and select the corresponding replicate as the dependent variable.

For each analysis, select Main Effects, enable Means and Signal to Noise Ratios, and select Smaller is better. Use Temp, Borex, pH, and Time as the Control Factors and Noise_Factor as the Noise Factor. Click Execute to perform each analysis. Then, click Show Chart to visualize the results.

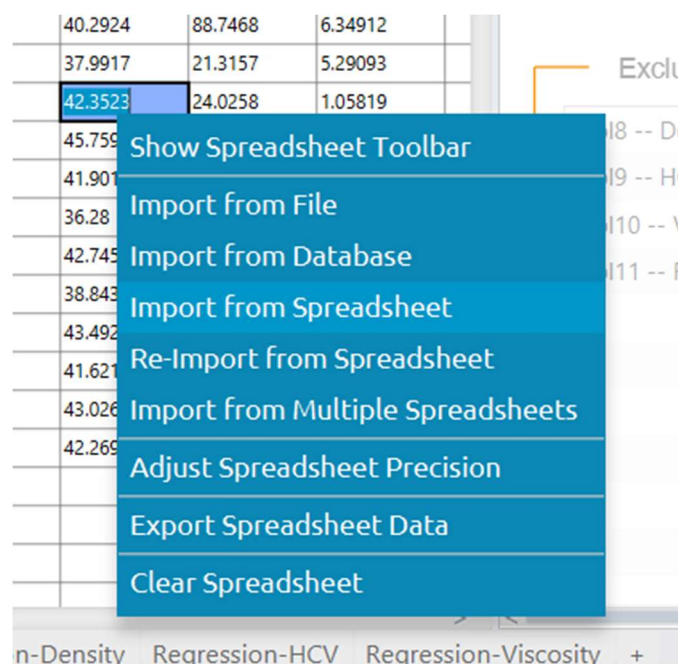
Note: Isalos allows only one dependent variable to be selected at a time in the Post-Robust Parameter Analysis. Therefore, the three experimental replicates (R1, R2, and R3) were analysed separately.

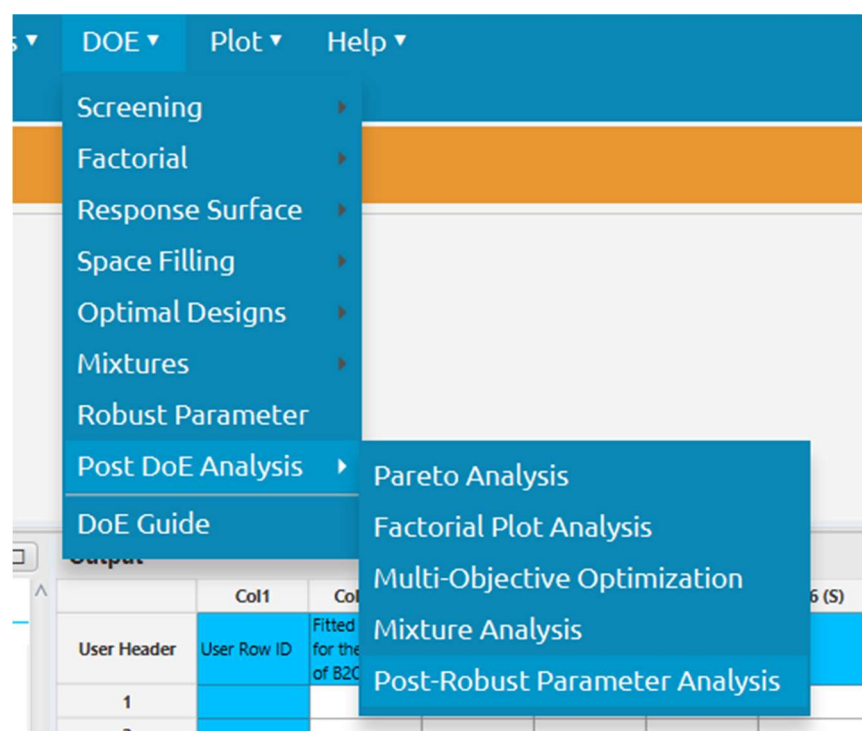
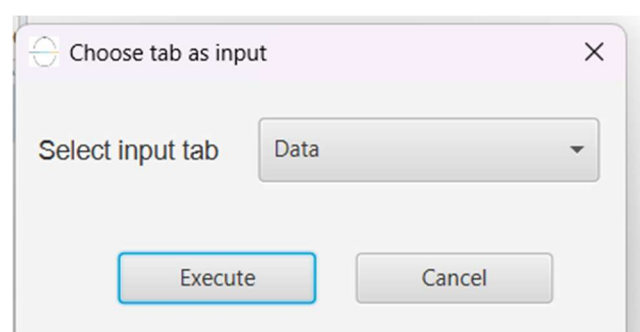
Step 6: B2O3_R1 Post-Robust Parameter Analysis

Create a new spreadsheet and rename it “B2O3_R1 Taguchi”. Right-click on the spreadsheet, select Import from Spreadsheet, choose the Data tab as input, and click Execute.

Then, go to DOE → Post DOE Analysis → Post-Robust Parameter Analysis. Select B2O3_R1 as the dependent variable and Main Effects as the analysis type. Select Means and Signal to Noise Ratios, and choose Larger is better as the Signal-to-Noise Ratio type. Select Temp, Borex, pH, and Time as the Control Factors and Noise_Factor as the Noise Factor.

Click Execute to perform the analysis. Then, click Show Chart to visualize the Main Effects and Signal-to-Noise Ratio results.





Post-Robust Parameter Analysis

?

×

Dependent Variable

Col10 -- B2O3_R1

Analysis Type

Main Effects

Confidence Level (%)

95.0

Fit a model for

☒ Means
 ☐ Std Deviation
 ☐ Ln(Std Deviation)
 ☒ Signal to Noise Ratios

Signal to Noise Ratio Type

Larger is better

☐ Find Optimal Settings

Optimization Objective for Mean

Maximize

Excluded Columns

Col7 -- Lol_R1

Col8 -- Lol_R2

Col9 -- Lol_R3

Col11 -- B2O3_R2

Col12 -- B2O3_R3

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Control Factors

Col2 -- Temp

Col3 -- Borex

Col4 -- pH

Col5 -- Time

Specify Reference Levels

Noise Factors

Col6 -- Noise_Factor

Execute

Cancel

Results :

Output																				
	Col1	Col2 (S)	Col3 (S)	Col4 (S)	Col5 (S)	Col6 (S)	Col7 (S)	Col8 (S)	Col9 (S)	Col10 (S)	Col11	Col12 (S)	Col13 (S)	Col14 (S)	Col15 (S)	Col16 (S)	Col17 (S)	Col18 (S)	Col19	
User Header	User Row ID	Fitted Model for the Means of B2O3_R1										Fitted Model for the Signal To Noise Ratio of B2O3_R1								
1																				
2																				
3		Mean Model Response Table										Signal To Noise Ratio Model Response Table								
4		Level	Temp	Borex	pH	Time						Level	Temp	Borex	pH	Time				
5		1	38.113333	41.103333	39.48	42.913333						1	31.5347490	32.270089	31.8091046	32.6333624				
6		2	36.446667	38.656667	38.716667	34.633333						2	31.1772488	31.6348618	31.6343432	30.6896120				
7		3	44.76	39.56	41.123333	41.773333						3	33.0105538	31.8176006	32.2791038	32.3995773				
8		Delta	8.313333	2.446667	2.406667	8.280000						Delta	1.8333050	0.6352276	0.6447606	1.9437504				
9		Rank	1.0	3.0	4.0	2.0						Rank	2.0	4.0	3.0	1.0				
10																				
11		Mean Model ANOVA Table										Signal To Noise Ratio Model ANOVA Table								
12		Source	DF	Adj SS	Adj MS	F-Value	P-Value	Significant				Source	DF	Adj SS	Adj MS	F-Value	P-Value	Significant		
13		Temp	2	116.0674667	58.037333							Temp	2	5.6668135	2.8334068					
14		Borex	2	9.1840667	4.592033							Borex	2	0.6416536	0.3208268					
15		pH	2	9.0752667	4.537633							pH	2	0.6671571	0.3335785					
16		Time	2	120.8376	60.4188							Time	2	6.7568023	3.3784011					
17		Error	0									Error	0							
18		Total	8	255.1644000								Total	8	13.7324265						
21																				
22		Mean Model Coefficients Table										Signal To Noise Ratio Model Coefficients Table								
23		Variable	Coefficient	Std. Error	Lower CI	Upper CI	Test Statistic	df	P-Value	Significant		Variable	Coefficient	Std. Error	Lower CI	Upper CI	Test Statistic	df	P-Value	Significant
24		Intercept	39.773333	0.333333	39.1200120	40.4266547		1.0				Intercept	31.9075172	0.333333	31.2541959	32.5608385		1.0		
25		Temp_600	-1.660000	0.4714045	-2.5839359	-0.7360641		1.0				Temp_600	-0.3727682	0.4714045	-1.2967041	0.5511677		1.0		
26		Temp_800	-3.326667	0.4714045	-4.2506025	-2.4027308		1.0				Temp_800	-0.7302684	0.4714045	-1.6542043	0.1936675		1.0		
27		Borex_10	1.330000	0.4714045	0.4060641	2.2539359		1.0				Borex_10	0.3625721	0.4714045	-0.5613638	1.2865080		1.0		
28		Borex_30	-1.116667	0.4714045	-2.0406025	-0.1927308		1.0				Borex_30	-0.2726555	0.4714045	-1.1965913	0.6512804		1.0		
29		pH_15	-0.293333	0.4714045	-1.2172692	0.6306025		1.0				pH_15	-0.0984126	0.4714045	-1.0223485	0.8255233		1.0		
30		pH_25	-1.056667	0.4714045	-1.9806025	-0.1327308		1.0				pH_25	-0.2731740	0.4714045	-1.1971099	0.6507619		1.0		
31		Time_60	3.140000	0.4714045	2.2105641	4.0693939		1.0				Time_60	0.7258452	0.4714045	-0.1980907	1.6487910		1.0		
32		Time_100	-5.140000	0.4714045	-6.0693939	-4.2105641		1.0				Time_100	-1.2179052	0.4714045	-2.1416471	-0.2939693		1.0		
33																				

Step 7: B2O3 Replicate Analysis

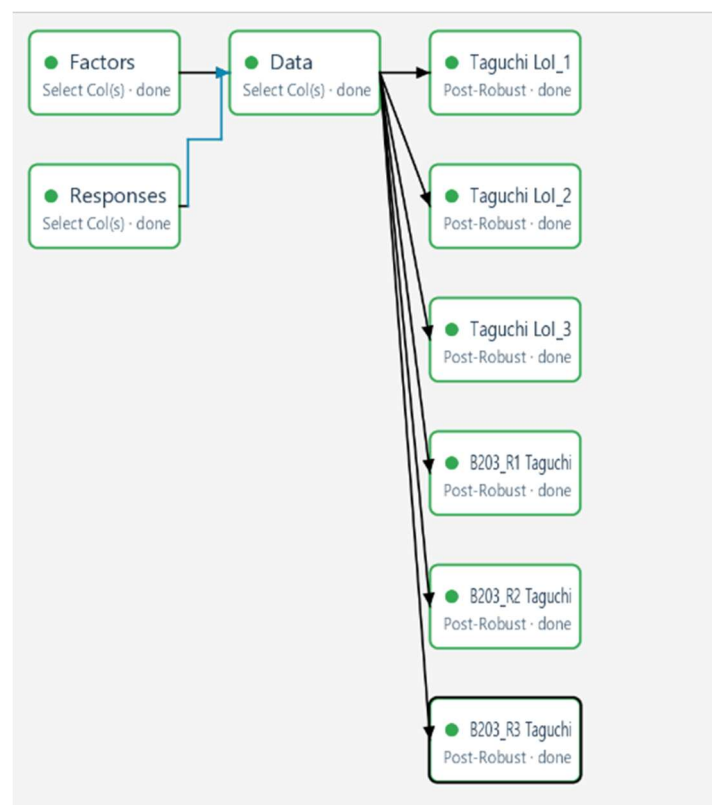
Repeat the Post-Robust Parameter Analysis for the remaining B2O3 replicates, B2O3_R2 and B2O3_R3. Create a separate spreadsheet for each replicate, import the Data spreadsheet, and select the corresponding replicate as the dependent variable.

For each analysis, select Main Effects, enable Means and Signal to Noise Ratios, and choose Larger is better. Use Temp, Borex, pH, and Time as the Control Factors and Noise_Factor as the Noise Factor. Click Execute, and then click Show Chart to visualize the results.

Note: As Isalos allows only one dependent variable to be selected at a time, B2O3_R2 and B2O3_R3 were analysed separately, following the same procedure as B2O3_R1.

Final Isalos Workflow

The final workflow is presented below :



References

Patyal, V. S., Modgil, S., & Maddulety, K. (2013). *Application of Taguchi Method of Experimental Design for Chemical Process Optimisation: A Case Study*. **Asia-Pacific Journal of Management Research and Innovation**, 9(3), 231–238. <https://doi.org/10.1177/2319510X13519320>